

MANIFEST

BOB-094 — Tracker-fetch stress + chaos automation evidence

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Scope

tests/stress/test_tracker_fetch_stress_chaos.py — §11.4.85 stress + chaos coverage for SearchOrchestrator.fetch_torrent (the outbound HTTP + response classification path in download-proxy/src/merge_service/search.py).

Scenarios delivered (8 + 1 meta = 9 tests, all GREEN)

STRESS

#	Test	Category
1	test_stress_sustained_100_sequential_fetches	sustained-load — p50/p95/p99 over 100 iters
2	test_stress_concurrent_10_parallel_no_fd_leak	concurrent-contention — no deadlock, no fd leak
3	test_boundary_empty_url_returns_none_or_error	boundary — empty URL
4	test_boundary_max_length_url_survives	boundary — 2000-char URL

#	Test	Category
5	test_boundary_off_by_one_content_type_edges	boundary — CT classifier edges

CHAOS

#	Test	Category
6	test_chaos_network_drop_20pct_categorised_as_network	network-fault — §11.4.69 feature_class=net
7	test_chaos_midflight_kill_clean_degradation	process-death — HTTP server killed mid-fetch
8	test_chaos_input_corruption_malformed_response_quarantined	input-corruption — malformed body quarantined
9	test_section_114_85_category_map	meta — mechanic coverage-claim assertion

§11.4.115 RED capture (production-mutation-driven)

Scenarios with captured RED:

test_chaos_input_corruption_malformed_response_quarantined AND test_boundary_off_by_one_content_type_edges.

Applied always-accept mutation to the CT/prefix classifier at download-proxy/src/merge_service/search.py:1893-1898 (replaced the boolean guard with `if True:` so a malformed HTML body would be returned to the caller). Both tests FAILED with user-observable assertion errors proving the classifier bypass was detected — see evidence/red_capture_against_mutated_classifier.txt.

Production tree restored (`git diff --stat` empty); re-ran the full suite — 9/9 GREEN — see evidence/green_capture_after_restore.txt.

Evidence artefacts

evidence/*.json — one per scenario, written by the tests themselves via the Python analogue of `ab_pass_with_evidence` (path exists + non-empty + parses). Each carries `section=11.4.85`,

feature_class per §11.4.69 taxonomy, and user-observable metrics (latency percentiles, byte-equality, FD deltas, error-category counts, quarantine counts).

Discipline

- §11.4.161 rootless — no root, hermetic (in-process aiohttp stub + ephemeral loopback HTTP server for the mid-flight-kill scenario)
- §11.4.5 captured evidence — per-iteration latency histogram, categorised error counts, state deltas
- §11.4.14 chaos cleanup — atexit + finally teardown for the real HTTP server (no bound port left behind)
- §11.4.69 taxonomy — feature_class recorded per scenario
- §11.4.201 no false-null — FD-leak check honestly reports unmeasurable_on_this_host when /proc/<pid>/fd is unavailable
- §11.4.113 — no force-push in this delivery